Feasibility of Using Large Language Models for Requirement-Based Software Testing

Alexander Seyr

Outline

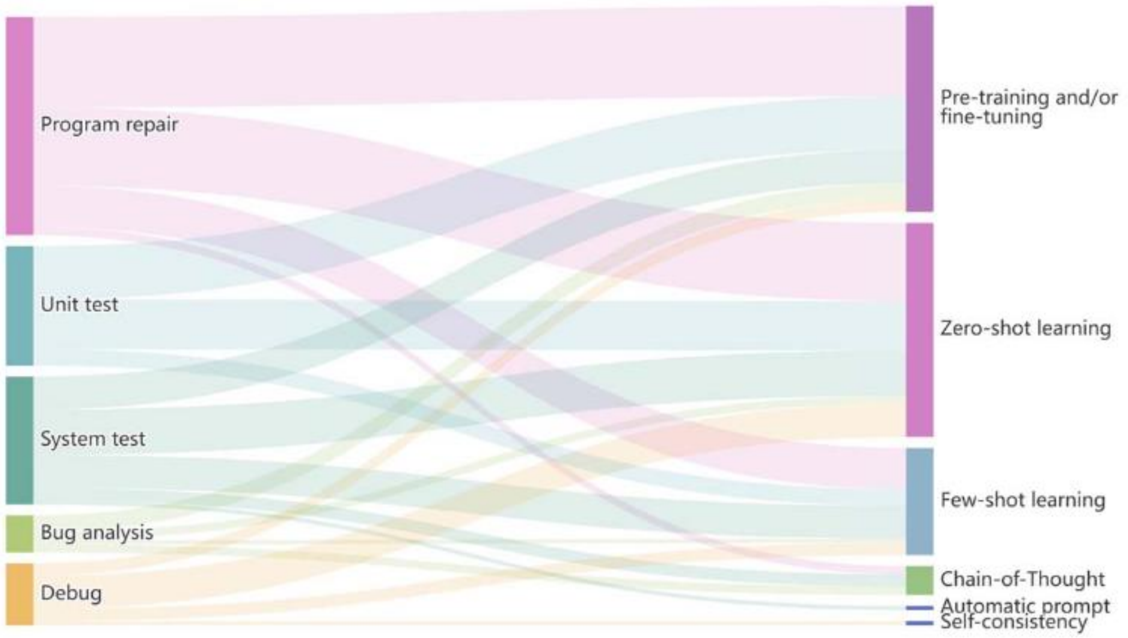
- Software testing overview
- Focus of this project
- Experimental Setup
  - Evaluation Procedure
  - Dronology Dataset
- Prompt engineering techniques
- Models
- Results – discussion & challenges

Software testing overview

- Quality assurance through testing

- Strategies
  - Unit testing
  - Integration testing
  - Acceptance/Validation testing
  - System testing

- Research on applying LLM’s for different testing strategies:

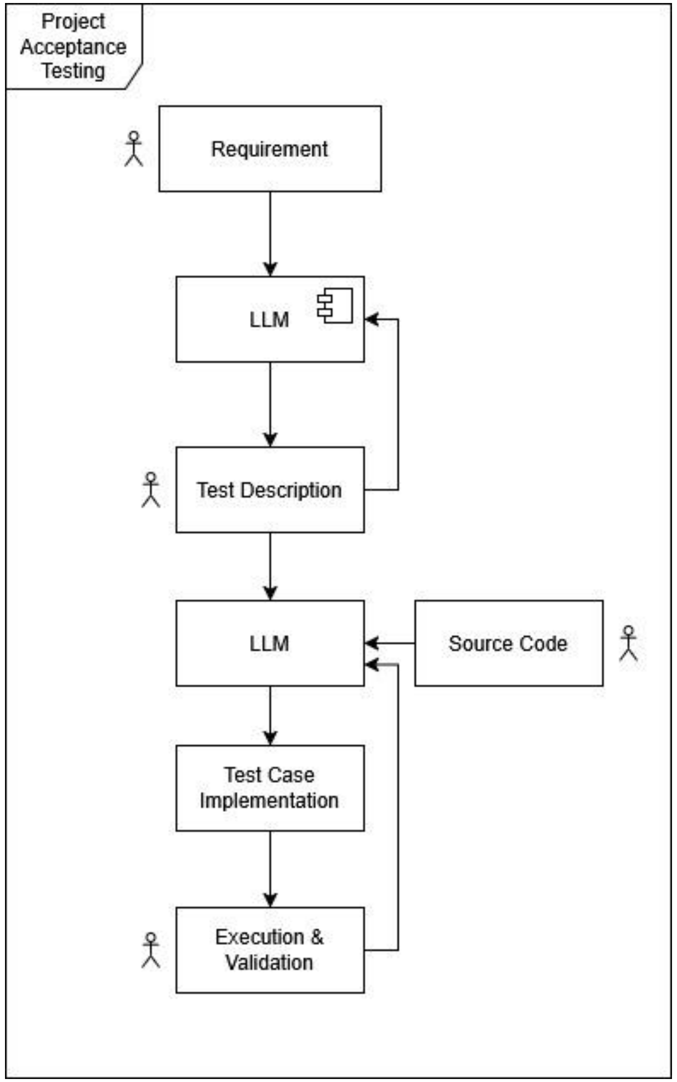


Wang, Junjie / Huang, Yuchao / Chen, Chunyang / Liu, Zhe / Wang, Song / Wang, Qing, **Software testing with large language models: Survey, landscape, and vision** 2024 *IEEE Transactions on Software Engineering*
Anwar, Nahid / Kar, Susmita, **Review paper on various software testing techniques & strategies,** 2019, *Global Journal of Computer Science and Technology* , Vol. 19, No. 2 p. 43-49

Focus of this study – Acceptance Testing

- Are requirements fullfilled?

- How capable are LLM’s in:
    - Understanding requirements in technical domain?
    - Deriving test descriptions from requirements?



Experimental Setup

- Evaluation Procedure
  - Use prompt engineering techniques to generate testcases
  - Validation structure for initial quality estimate
    - Model + prompt variations
  - Basic Test Description structure shall be enforced

```
{
    "requirement": "",
    "testObjective": "",
    "preconditions": [],
    "testSteps": [],
    "expectedResult": ""
},
```

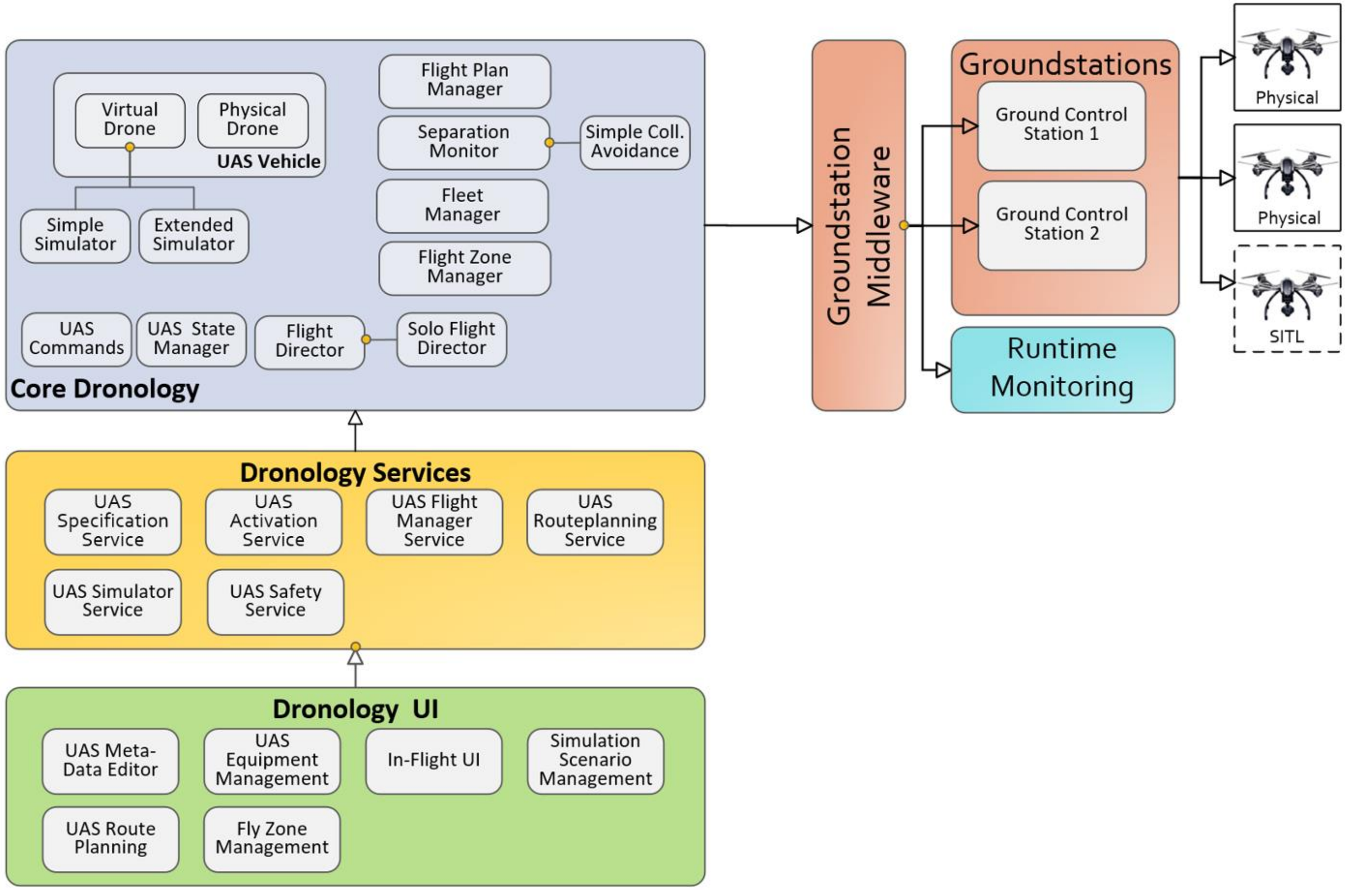
  - Human verification as final evaluation

Experimental Setup

- Evaluation on Dronology

  Unmanned Aerial Systems Management and Control

  - research environment
  - framework for controlling and coordinating the flight
  - https://dronology.info/



- 97 Requirements

RE-8 -- UAV State transitions	[Requirement]
Status: Closed	
Description: When requested the _VehicleCore_ shall shall transition the UAV between states according to allowed state transitions as depicted in the UAV state transition diagram	

RE-741 -- Message frequency assignment	[Requirement]
Status: Closed	
Description: The _GCS_ shall assign a message frequency for all UAVs.	

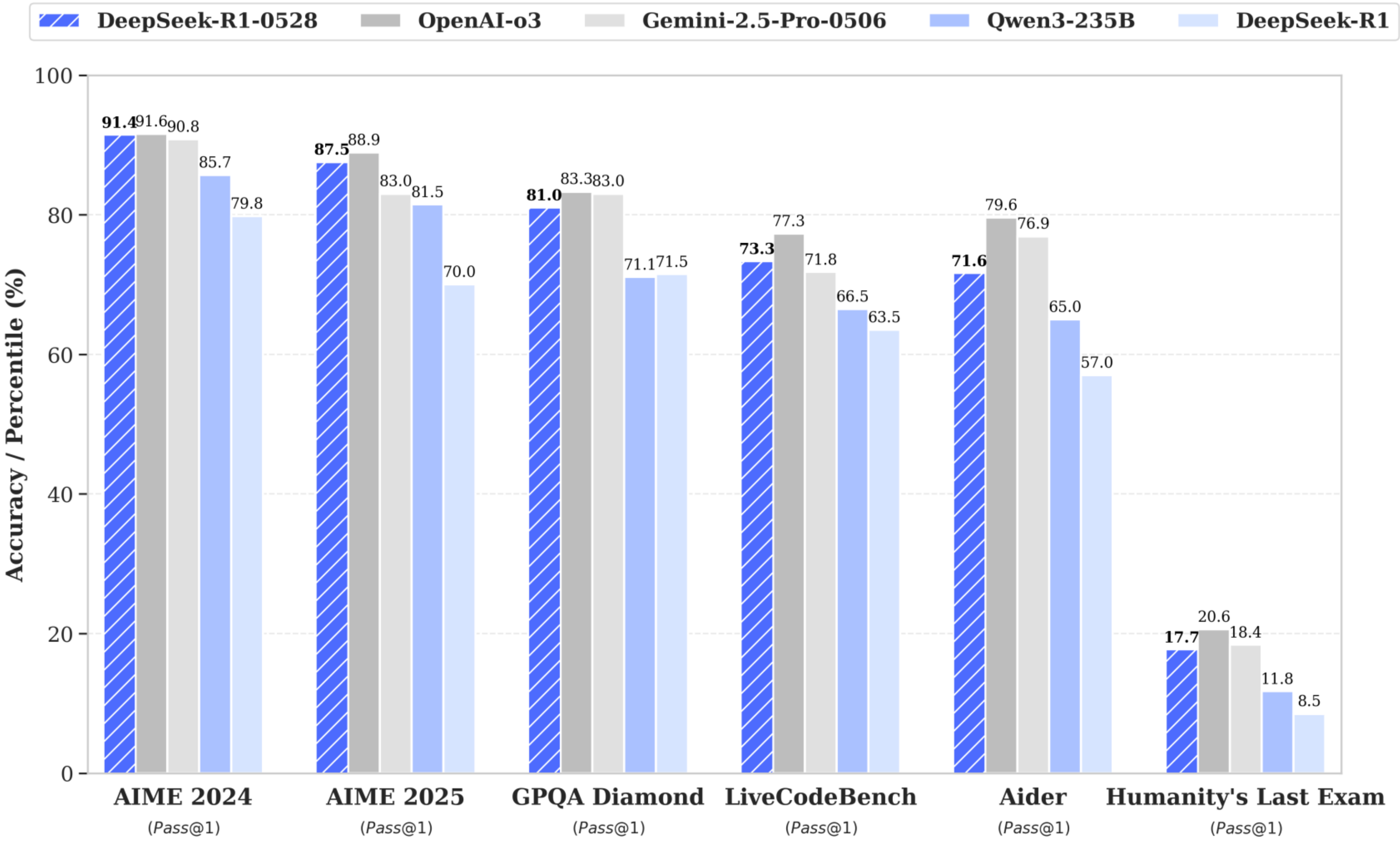
RE-766 -- Multiple Map Types	[Requirement]
Status: Closed	
Description: The _MapComponent_ shall support different types of map layers (e.g. terrain satellite)	

Prompt Engineering techniques

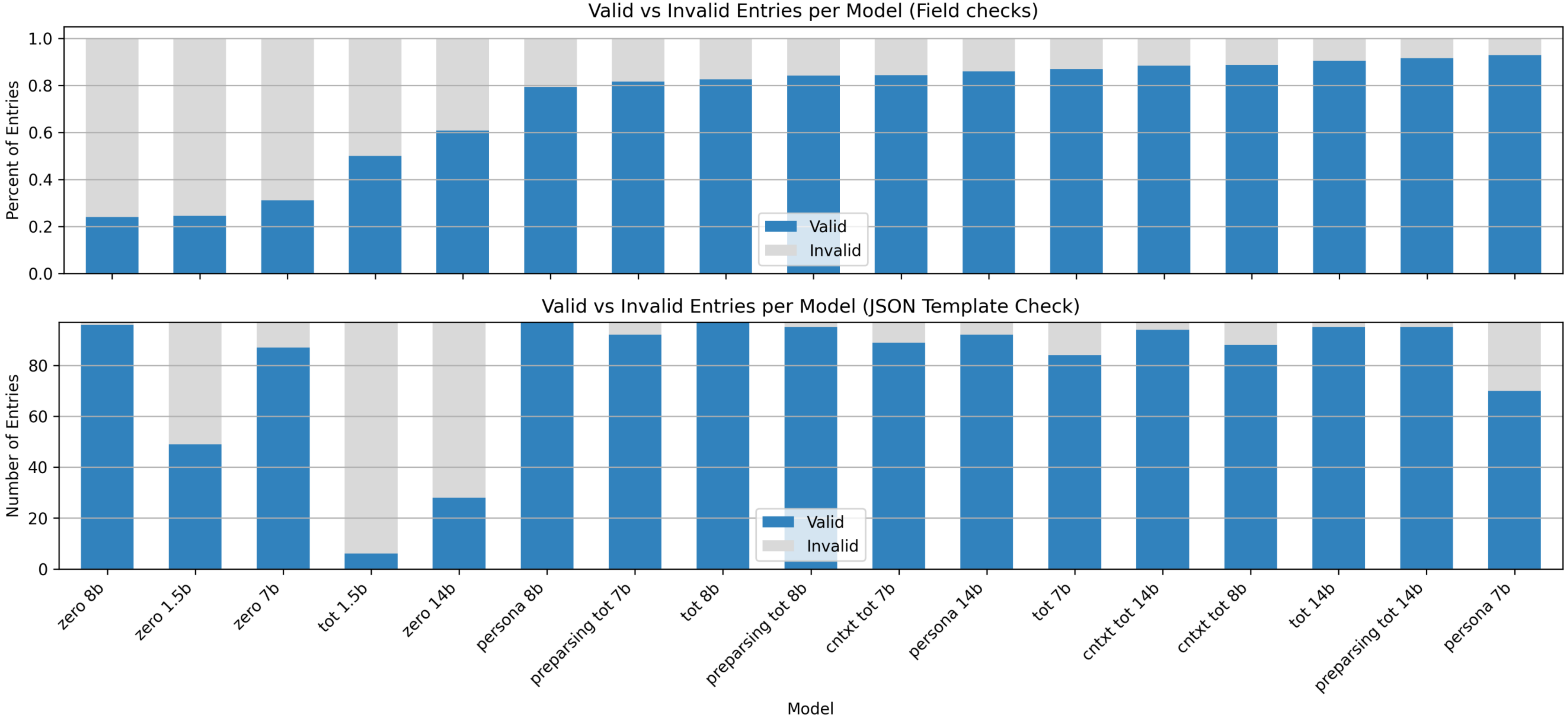
1. Zero shot learning
   \+ Directional stimulus
   Baseline
2. Personas – software tester
3. Tree of thought – multiple experts
4. Providing Context
   - Dronology introduction
   - Preparsing of requirements + TOT

Models - Deepseek-r1

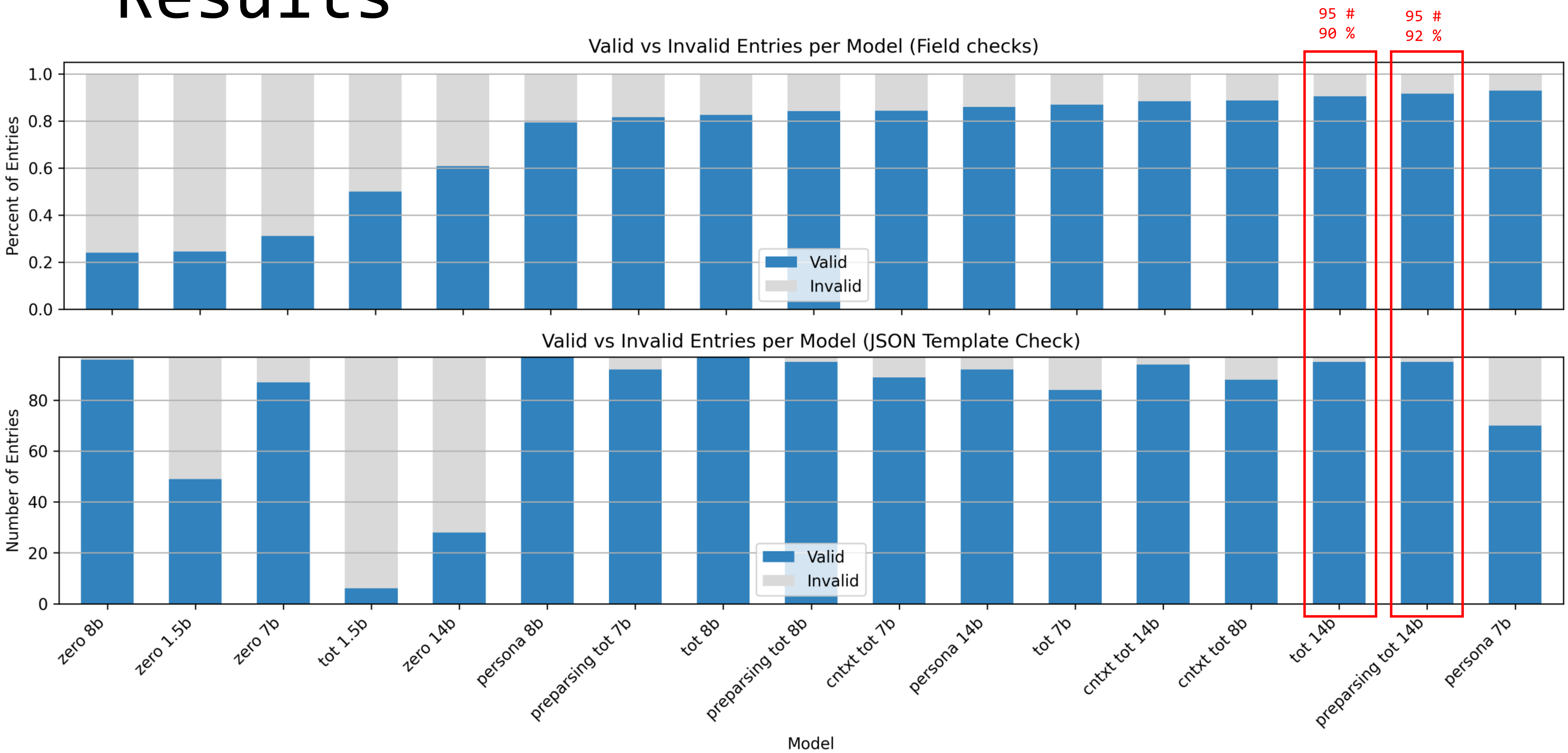
- Distilled
  - 1.5b
  - 7b
  - 8b
  - 14b
  - 32b
  - …
- Ollama
  - local
  - independent
- Fixed Parameters



Results


Valid vs Invalid Entries per Model (Field checks)
Percent of Entries
1.0
0.8
0.6
0.4
0.2
0.0
Valid
Invalid
Valid vs Invalid Entries per Model (JSON Template Check)
Number of Entries
80
60
40
20
0
Valid
Invalid
zero 8b
zero 1.5b
zero 7b
tot 1.5b
zero 14b
persona 8b
preparsing tot 7b
tot 8b
preparsing tot 8b
cntxt tot 7b
persona 14b
tot 7b
cntxt tot 14b
cntxt tot 8b
tot 14b
preparsing tot 14b
persona 7b
Model

Results


95 #
90 %
95 #
92 %
Valid vs Invalid Entries per Model (Field checks)
Percent of Entries
1.0
0.8
0.6
0.4
0.2
0.0
Valid
Invalid
Valid vs Invalid Entries per Model (JSON Template Check)
Number of Entries
80
60
40
20
0
Valid
Invalid
zero 8b
zero 1.5b
zero 7b
tot 1.5b
zero 14b
persona 8b
preparsing tot 7b
tot 8b
preparsing tot 8b
cntxt tot 7b
persona 14b
tot 7b
cntxt tot 14b
cntxt tot 8b
tot 14b
preparsing tot 14b
persona 7b
Model

Q & A

Challenges

- Output generation
  - Unstable output with default parameters
    - Increased context window
    - Decreased repeat penalty
    - Decreased top_p
  - 1.5b – very unreliable
  - >32b parameters - local resource limit
- Field Rules
  - Present tense check uncertanties

Field check analysis

- Example output - persona prompt 14b


```
"Requirement": "The RouteCreationUI shall allow the user to create and modify a route.",
"testObjective": "Verify that the RouteCreationUI allows the user to create and modify a route.",
"preconditions": [
    "The application is running.",
    "The user has the necessary permissions to create and modify routes."
],
"testSteps": [
    "1. Launch the application.",
    "2. Navigate to the RouteCreationUI.",
    "3. Click on the 'Create New Route' button to initiate creating a route.",
    "4. Input the necessary route details such as waypoints, altitude, speed, etc.",
    "5. After entering all required details, click 'Submit' to create the route.",
    "6. Verify that the route is successfully created and appears in the route list.",
    "7. Modify an existing route by navigating to an already created route in the list.",
    "8. Make changes to the route details such as updating waypoints, altitude, or speed.",
    "9. Click 'Update' to save the changes.",
    "10. Verify that the route has been modified with the new details."
],
"expectedResult": "The user can successfully create and modify a route through the RouteCreationUI, and the changes are reflected in the system.",
"testCaseID": "RE-672",
```